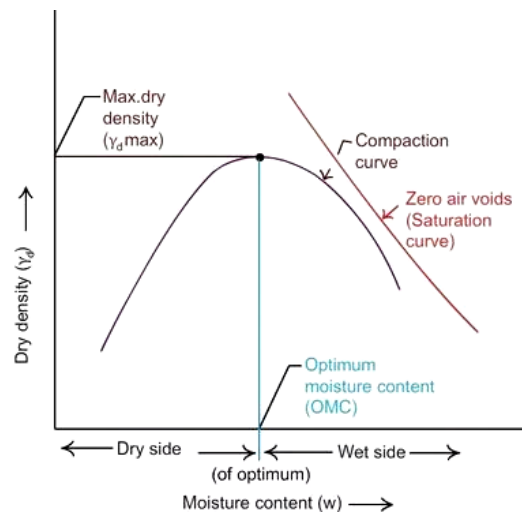


Experiment	Standard Proctor Compaction Test
Objective	To determine the maximum dry density and the optimum moisture content of a soil sample
Apparatus	Proctor mould with the detachable collar assembly and the base plate (volume of the mould is 1000 cm³) Manual rammer of 2.5 kg weight which is equipped to provide a standard drop from a height of 30 cm Sample Extruder Electronic balance which can be readable to 0.001 kg Straight edge Mixing tools such as mixing pan, spoon, trowel, spatula etc. Moisture cans and a drying oven A squeeze bottle or a beaker of water

Theory

Compaction is the process of densification of soil by reducing air voids. The degree of compaction of a soil is measured in terms of its dry density. The Proctor compaction test is used to determine the Optimum Moisture Content (OMC) which gives the Maximum Dry Density (MDD) of the compacted soil.

The maximum dry density and optimum moisture content can be found by the graph of dry density verses moisture content.



Procedure

1. Obtain approximately a 3 kg sample of soil which is passed through 20 mm sieve.
2. Record the weight of the Proctor mould with the base and without the collar extension, the volume of which is 1000 cm³.
3. Assemble the compaction apparatus.
4. Add adequate amount of water to the sample and mix thoroughly.
5. Fill the mould with wet soil in 3 layers and compact them by applying 25 well distributed blows from Proctor hammer.
6. Detach the collar without disturbing the soil inside the mould.
7. Level the top surface of the soil sample using a knife edge.
8. Determine the weight of the compacted wet soil sample with the mould and the base.
9. Remove the compacted soil from the mould and take a sample (50 g -100 g) of soil to obtain the water content. Keep them in the oven for 24 hours approximately.
10. Place the remaining soil inside the mould into the pan, break it down and remix it thoroughly with the other soil.
11. Add suitable amount of extra water & mix them well.
12. Repeat the above procedure. (During this process the mould+wet soil weight will increase with the increase in soil moisture and then drops gradually) Take two moisture increments after achieving the maximum weight. Obtain at least 4 points to plot the dry density with variation of moisture content.
13. After 24 hours, recover the samples in the oven and determine the dry weights of them.

Calculation

1. Compute the bulk density and moisture content of each compacted specimen.
2. Compute the dry density of each compacted specimen using the following equation;
$$\gamma_{dry} = \frac{\gamma_{bulk}}{(1 + \omega)}$$

γ_{dry} = Dry density of the compacted soil
 γ_{bulk} = Bulk density of the compacted soil
 ω = Water content
3. Plot dry density against the corresponding moisture content.
4. Draw smooth curve through the resulting points.

Results Obtain the maximum dry density and the optimum moisture content from the graph.

Conclusion How are the above results used in field compaction.

OBSERVATIONS

Proctor Compaction Test

Name

Reg Number

Course

Date

Trial Number	Volume of Mould (cm ³)	Mass of Mould (kg)	Mass of Mould+ Soil (kg)	Moisture can No	Mass of Can (g)	Mass of Can + Wet soil (g)	Mass of Can + Dry soil (g)